

Why Slovene Refusal Outlasts English Abliteration: An Internal Research Record

AI Inventor

September 2026

Abstract

GaMS3-12B-Instruct and Gemma-3-12B-IT share a pretrained ancestor but diverge in safety supervision language. Five iterations of experiments test whether Slovene-taught refusal transfers to English, whether English-only ablation removes Slovene refusal equally in both models, and what mechanism explains the residual Slovene refusal. The cross-lingual ablation lag is confirmed across every readout, judge, and item body: at matched English refusal, GaMS3 retains less Slovene refusal than Gemma. The lag is baseline, not edit-induced (R-BASE). The lag is better characterised as an output-language effect (C-OUT) than a generic cross-lingual lag: experiment 16 replicates the finding on a fresh 294-item body with prediction-powered inference correction (output-language contrast for Gemma = +2.43 [+0.89, +5.38] log-odds; cross-model difference = -2.31 [-5.34, -0.68]). Gemma’s Slovene-output reserve is real safety, not production failure: harmful content separates at both doses (harmful-content output-language contrast = 2.61 [2.20, 3.07] at high dose), and the four-class decomposition shows half of the non-compliance is deflection (production-failure index difference 0.46 [0.23, 0.65]), consistent with Shen et al.’s relevance curse. On RefusEU FINAL prompts ($n = 1,300$ per cell), the edit is effective in all four cells (ROBUST). The novelty verdict is downgraded from NOVEL to PARTIALLY OCCUPIED after Addagada (2609.08373) and Nguyen (2608.26186). The readout gate has never passed; all adjudication is author-LLM, not human.

Contents

1	Framing	4
2	Iteration 1	4
2.1	Strategy	4
2.2	Artifact 1: Dataset and supervision audit	5
2.3	Artifact 2: Main screen of refusal differences (Experiment 1)	6
2.3.1	Refusal rates	6
2.3.2	Difference-in-differences and dose contrast	6
2.3.3	Item-matched machine-translation arm	8
2.3.4	Identity calibration (C1)	8
2.3.5	Surface forms	8
2.3.6	Verification	8
2.4	Artifact 3: Base models and mechanistic induction (Experiment 2)	8
2.4.1	Alternate 2 (base-checkpoint geometry): killed	8
2.4.2	C4 (ancestor-direction induction): failure branch	9
2.5	Artifact 4: Persona-gate test (Experiment 3)	9

2.5.1	Verdict: untestable	9
2.6	Artifact 5: Heretic ablation and transfer curve (Experiment 4)	10
2.6.1	English ablation reaches Slovene	10
2.6.2	C3 (frozen-core transfer): fails both ways	10
2.6.3	C5a (excess Slovene KL): reverses the prediction	10
2.6.4	C5b (bilingual re-selection): unpowered	10
2.6.5	Alternate 4 (thin templated gate): strongest signal but does not survive . . .	10
2.6.6	Failed attempts and limits	12
2.7	Summary of dead ends and surviving claims after Iteration 1	12
3	Iteration 2	12
3.1	Strategy	12
3.2	Artifact 6: Confirmation on the reserved FINAL splits (Experiment 5)	13
3.2.1	Identity calibration: passes on FINAL	13
3.2.2	Refusal on RefusEU natural pairs	13
3.2.3	HARD set: signal-detection analysis	13
3.2.4	Refusal style	14
3.3	Artifact 7: Teacher-inheritance screen (Experiment 7)	15
3.3.1	Decision inheritance: uninterpretable	15
3.3.2	Wording inheritance: supported	15
3.4	Artifact 8: Heretic transfer-curve re-test (Experiment 8)	15
3.5	Artifact 9: Reconciliation of iteration-1 findings (Evaluation 1)	17
3.6	Artifact: Refusal-depth pipeline (Experiment 6)	17
4	Iteration 3	17
4.1	Strategy	17
4.2	Artifact 10: Screen repair of experiment 8 (Evaluation 2)	19
4.3	Artifact 11: Cross-lingual ablation lag confirmation (Experiment 9)	19
4.4	Artifact 12: Mechanism test (Experiment 10)	19
4.4.1	Geometry: EN and SL refusal directions are nearly collinear	21
4.4.2	Induction: the Slovene-perpendicular direction does not induce refusal	21
4.4.3	Add-on neutralisation	21
4.5	Artifact 13: FINAL confirmation on reserved split (Experiment 11)	21
4.5.1	Hungarian arm	21
4.5.2	Belebele comprehension: unchanged by the edit	22
4.5.3	Mixed-effects model	22
4.6	Artifact 14: Utility control (Experiment 12)	22
4.7	Not executed or abandoned in iteration 3	23
5	Iteration 4	23
5.1	Strategy	23
5.2	Artifact 15: Readout completion with paid judge (Evaluation 3)	24
5.2.1	Readout gate: FAILS	24
5.2.2	Pooled cross-model lag	24
5.3	Artifact 16: $2 \times 3 \times 3$ mechanism test (Experiment 14)	25
5.4	Artifact 17: Fine-grained lambda ladder (Experiment 15)	25
5.5	Artifact 18: Prior-art and fact check (Research)	27
5.6	Not executed or abandoned in iteration 4	28

6	Iteration 5	28
6.1	Strategy	28
6.2	Artifact 19: C-OUT confirmation (Experiment 16)	28
6.3	Artifact 20: RefusEU FINAL (Experiment 17)	28
6.4	Artifact 21: R-INCAP safety evaluation (Evaluation 4)	31
6.5	Artifact 22: Paper number audit (Evaluation 5)	31
6.6	Artifact 23: Novelty and fact check (Research 2)	31
6.7	Not executed or abandoned in iteration 5	32
7	What We Have Learned	32
8	Run Bookkeeping	33

1 Framing

GaMS3-12B-Instruct [20] and Gemma-3-12B-IT both descend from `google/gemma-3-12b-pt`, but followed different post-training paths. GaMS3 received Slovene continued pretraining on 140 billion tokens and supervised fine-tuning on a mix that is roughly 79% Slovene, including about 7,000 safety-refusal examples drawn largely from machine-translated Qwen-family responses. Gemma-IT received Google’s standard multilingual post-training. Because the two checkpoints share a pretrained ancestor yet diverge in the language of their safety supervision, they form a natural pair for testing whether the language in which refusal was taught determines where refusal appears.

The core question is whether Slovene-taught refusal transfers to English. Prior work on refusal directions [1] established that a single subspace mediates refusal in English, and Wang et al. [2] showed that English-extracted refusal directions transfer across 14 languages, but both studies used models whose post-training mixed many languages. Less is known about transfer when safety supervision is concentrated in one language. Shaham et al. [4] found that “a pinch” of multilingual instruction data suffices for task-level transfer; Krasnodebska et al. [8] found that English-only DPO alignment does *not* ensure cross-lingual safety, even for the same harm categories. This tension motivates the dose-response design: if the English share of GaMS3’s refusal supervision varies across hazard categories, categories with essentially zero English examples should show a measurable Slovene surplus that Gemma does not share.

A second question follows: if refusal does transfer, does the English-only objective of an automatic ablation tool (Heretic [24]) remove Slovene refusal as readily as English refusal, and does it do so equally in GaMS3 and Gemma-IT? The practical stake is whether a monolingual de-censoring pipeline is sufficient to strip safety from a bilingual model, or whether the Slovene supervision gives GaMS3 a Slovene reserve that an English-only adversary cannot reach. This framing serves *evaluation and defence*: measuring the cross-lingual reach of an English-only edit tells builders where their models remain protected and where they do not. Joad et al. [5] showed that category-specific refusal directions are geometrically distinct yet collapse under linear intervention; Zhong and Li [6] argued that refusal lives downstream of persona in chat models. Both findings bear on whether a single English-extracted direction is the right probe for a bilingual model.

Four rival accounts were pre-registered. The main claim (behaviour-level reach) predicts that refusal transfers to English at every hazard-category grain. Alternate 1 (supervised reach) predicts a Slovene surplus in low-English-dose categories. Alternate 2 (base-checkpoint geometry) predicts that GaMS-vs-Gemma differences are explained by the base models’ harmfulness separability [3, 14]. Alternate 3 (persona gating) predicts a uniform Slovene surplus driven by GaMS3’s Slovene-anchored self-identity [6]. Alternate 4 (thin templated gate) predicts symmetric language differences driven by the diversity, not the language, of refusal supervision.

2 Iteration 1

2.1 Strategy

The iteration ran five artifacts: a dataset artifact that built the evaluation sets and audited GaMS3’s supervision dose, and four experiment artifacts that each tested one or more of the pre-registered claims. The design was a screen: every claim was evaluated on SCORE items drawn from RefusEU’s train and test splits, with the reserved RefusEU evaluation split untouched for a future FINAL pass. The sequence was dictated by a single-GPU constraint (40 GB disk), requiring models to be loaded and evicted one at a time.

The four experiments were:

1. Experiment 1 (main screen): the refusal difference-in-differences between GaMS3 and Gemma-IT on original models, plus the dose contrast D and identity calibration.
2. Experiment 2 (base models and mechanistic): harmfulness separability in the two base checkpoints and two instruct models, plus the ancestor-direction induction test.
3. Experiment 3 (persona gate): the alternate-3 manipulation, ablating GaMS3’s persona direction and testing whether it moves Slovene refusal.
4. Experiment 4 (Heretic ablation): English-objective ablation of both instruct checkpoints, the transfer curve, the excess KL ratio, and the prefill test.

All protocols were hashed and committed before scoring. The OpenRouter shared key hit its daily limit partway through experiments 3 and 4, leaving some judge calls incomplete; these gaps are documented below.

2.2 Artifact 1: Dataset and supervision audit

Workspace: `gen_art_dataset_1`.

The dataset artifact assembled eight data blocks totalling 15,647 rows.

RefusEU evaluation split. 1,400 English and 1,400 Slovene prompts from RefusEU [8], frozen into 100 DEV and 1,300 FINAL pairs by sha1 of the pair identifier. Each item received a blind hazard-category label (majority vote of gpt-4.1-mini, gemini-2.5-flash, and a TF-IDF classifier). Gold-test accuracy was 0.759 for English and 0.716 for Slovene; restricting to items where all labellers agreed raised accuracy to 0.95 and 0.94. Inter-annotator kappa between the two LLM labellers was 0.88 (English) and 0.83 (Slovene).

A critical discovery was that same-row-identifier English and Slovene items in RefusEU are not translations: LaBSE cosine similarity was 0.553, matching the similarity of same-category pairs (0.547) and far below the machine-translation baseline (0.914). Zero of 1,400 pairs graded as translation-equivalent. This means within-model Slovene-vs-English comparisons on matched row identifiers conflate prompt content with language; only the cross-model difference-in-differences and a separate machine-translation arm are interpretable.

Supervision dose audit. 6,239 rows from GaMS-Nemotron-Chat and GaMS-Instruct-SAFE were labelled for response type, language, and hazard category by two LLM families (gpt-4.1-mini and gemini-2.5-flash, with llama-3.3-70b tiebreak on 928 disagreements; batch-check agreement 0.945 English, 0.959 Slovene). The overall English share of safety refusals was 0.178, with a propagated interval of [0.055, 0.452]. The upper bound exceeds 0.40, so “Slovene-dominant refusal” is not statistically licensed from the public data alone.

Table 1: Frozen dose groups after re-labelling (Artifact 1, Iteration 1).

Group	Categories	EN share range (point estimates)
Low-EN	S2, S5, S7, S13	0.034 to 0.096
High-EN	S1, S3, S4, S8, S9, S10, S11, S14	0.109 to 0.291
Intermediate	S6, S12	0.148 to 0.201

Thirteen of fourteen categories were flagged as dose-uncertain (the propagated EN-share interval straddles 0.10). The re-labelling changed S2 from high to low (point 0.075, down from Stage-0b’s 0.292) and S8 from low to high (point 0.226). S1 moved from intermediate to high (point 0.291).

Other blocks. The hard set (OR-Bench-hard-1k, XSTest, OR-Bench-toxic-300) was assembled with Gemini Slovene machine translation (back-translation chrF median 79; 11 items dropped). The identity set comprised 120 identity questions across six facets plus 120 matched personal controls, with zero overlap against the Nemotron training rows. The dataset artifact spent \$9.29 of a \$9.50 budget.

2.3 Artifact 2: Main screen of refusal differences (Experiment 1)

Workspace: `gen.art.experiment_1`.

Gemma-3-12B-IT and GaMS3-12B-Instruct were run in NF4 quantization with greedy decoding (64 new tokens, empty system prompt) on all 2,137 SCORE RefusEU pairs, a 400-pair item-matched machine-translation arm, 60 identity question pairs, and 1,504 CONSTRUCT pairs (for calibrating the continuous prefix score). Judges were gemini-2.5-flash on 10,852 responses and gpt-4.1-mini on a 20% validation subset; total judge spend was approximately \$0.91.

2.3.1 Refusal rates

The ceiling flag fired: Gemma’s English lexicon refusal rate was 0.961, and the prefix-score AUROC against the lexicon was only 0.61 in that cell, so the pre-registered fallback to the judge readout (including partial refusals) was triggered.

Table 2: Refusal rates on the 2,137 SCORE pairs (Experiment 1, Iteration 1).

Readout	Gemma EN	Gemma SL	GaMS EN	GaMS SL
Lexicon	0.961	0.885	0.942	0.723
Judge (binary)	0.979	0.929	0.971	0.865
Judge + partial	0.988	0.939	0.974	0.875

Both models refuse at high rates across languages. The gap is in Slovene: GaMS3 refuses less than Gemma in Slovene on both the lexicon (0.723 vs 0.885) and the judge (0.865 vs 0.929). In English, the two models are within two percentage points.

2.3.2 Difference-in-differences and dose contrast

The difference-in-differences ($\text{DiD} = [\text{GaMS}(\text{SL} - \text{EN})] - [\text{Gemma}(\text{SL} - \text{EN})]$) measures whether GaMS3 has a larger Slovene deficit than Gemma. A negative DiD means GaMS3’s Slovene gap is wider. All rate-to-logit conversions use the Hautus correction (adding 0.5 to counts before taking log-odds). The pre-registered threshold was $m = 0.40$ log-odds.

Table 3: Overall DiD and dose contrast D (log-odds, 95% bootstrap CI; Experiment 1, Iteration 1).

Readout	Overall DiD [95% CI]	D [95% CI]	MDE(D)
Lexicon	−0.67 [−0.95, −0.41]	0.61 [−0.58, 2.15]	1.89
Judge	−0.38 [−0.69, −0.05]	−0.08 [−1.27, 0.94]	1.53
Judge + partial	−0.03 [−0.39, 0.39]	−0.73 [−2.11, 0.48]	1.87

On the lexicon, the overall DiD is clearly negative (−0.67, CI excludes 0), confirming that GaMS3 has a wider Slovene refusal gap than Gemma. On the judge, the DiD is smaller (−0.38)

but its CI still excludes 0. Including partial refusals shrinks the DiD to near zero (-0.03), with the CI spanning both sides.

The dose contrast D (the difference in DiD between low-EN and high-EN categories) is the primary statistic for the main claim vs alternate 1. On all three readouts, D is uninformative: the confidence intervals are wide (MDE 1.5 to 1.9 log-odds), spanning both positive and negative values.

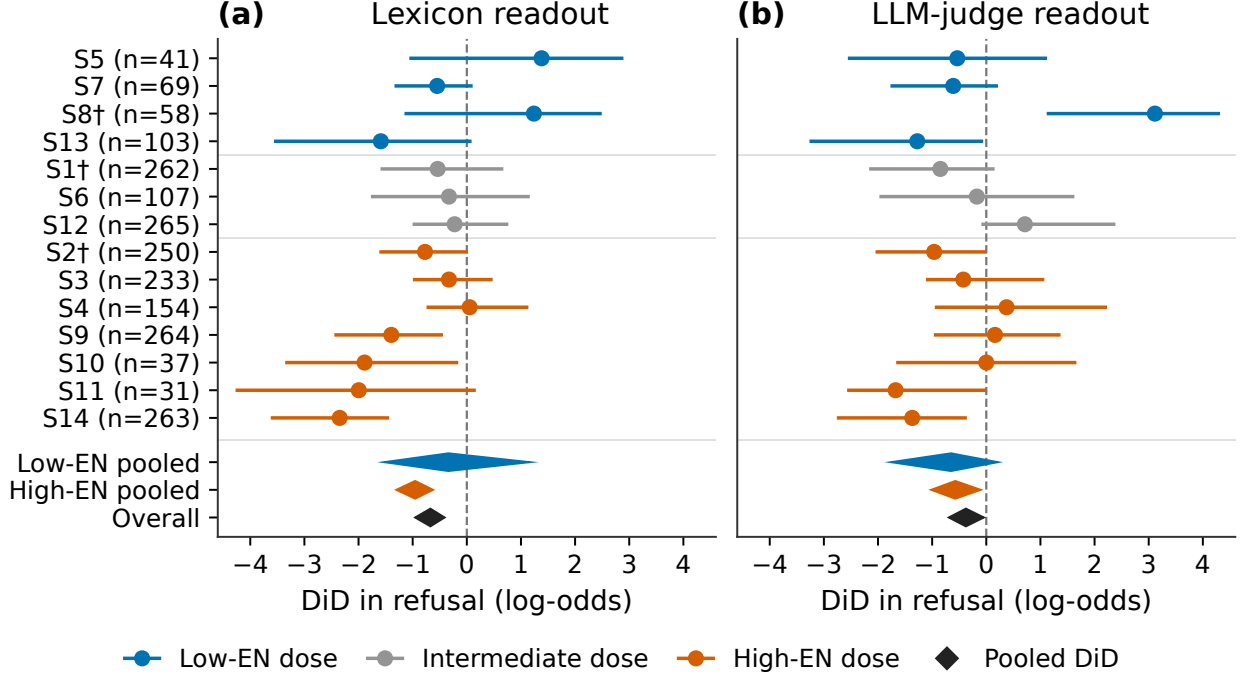


Figure 1: Difference-in-differences in refusal, $\text{DiD} = [\text{GaMS3}(\text{SL} - \text{EN})] - [\text{Gemma}(\text{SL} - \text{EN})]$ in log-odds, by hazard category on the 2,137 RefuseEU SCORE pairs of Experiment 1, for (a) the refusal lexicon and (b) the gemini-2.5-flash binary refusal judge. Negative values mean GaMS3’s Slovene refusal deficit is wider than Gemma’s. Circles are category estimates; horizontal lines are 95% pair-bootstrap CIs; the dashed line marks zero. Colour gives the English share of GaMS3’s safety-refusal supervision under the Stage-0b partition: low-EN (blue), intermediate (grey), high-EN (orange). Diamonds span the 95% CIs of the pooled low-EN, pooled high-EN and overall DiD. The overall DiD is -0.67 $[-0.95, -0.41]$ on the lexicon and -0.38 $[-0.69, -0.05]$ on the judge. The pooled low-EN and high-EN diamonds overlap on both readouts: $D = 0.61$ $[-0.58, 2.15]$ (lexicon) and -0.08 $[-1.27, 0.94]$ (judge).

The 14-category meta-regression of DiD on the English dose share found no significant slope. However, the item-level generalised linear mixed model produced a significant negative z_{EN} term on the lexicon (-0.17 , CI $[-0.28, -0.07]$) that flipped sign on the judge ($+0.13$, CI $[0.00, 0.27]$), a signature of pseudo-replication from 14 correlated category-level units (correlation between z_{EN} and z_{SL} : 0.86, VIF 3.84).

Scorer confound. The frozen Slovene lexicon lacked English-style “lecture” markers, missing 312 GaMS Slovene judge-classified refusals vs 97 for Gemma. Kappa between the lexicon and the judge was below 0.70 in three of four model-language cells.

2.3.3 Item-matched machine-translation arm

To separate the language effect from the prompt-content effect, a 400-item arm used NLLB machine translations of English prompts into Slovene. GaMS3’s Slovene-MT refusal rate was 0.968 vs its English rate of 0.970 (McNemar: 7 vs 8 discordant pairs). The judged DiD on this arm was -2.0 pp $[-4.3, 0.0]$, compared with -6.0 pp on the natural RefusEU pairs.

2.3.4 Identity calibration (C1)

The calibration arm passed. GaMS3 named itself “GaMS” in 90% of Slovene identity responses and 35% of English ones; Gemma named itself “Gemma” at 77% in English and 67% in Slovene. The identity DiD was 3.22 $[2.38, 4.38]$, far exceeding the 10 pp threshold ($m_{\text{id}} = 0.43$).

Table 4: Identity: own-name rates and name distribution (Experiment 1, Iteration 1).

Cell	Own-name rate	Top other names
GaMS EN	0.35	Qwen 0.13, none 0.48
GaMS SL	0.90	none 0.08
Gemma EN	0.77	none 0.22
Gemma SL	0.67	other 0.08, none 0.20

2.3.5 Surface forms

1,504 of 1,545 GaMS3 Slovene refusals (97.3%) opened with the calque “Oprostite, vendar...” (“I’m sorry, but...”), a word-for-word translation of the English Qwen refusal template.

2.3.6 Verification

The audit script passed 30 of 30 checks. Two independent re-derivation scripts reproduced all rates, DiD, D , DiD_{id} , and machine-translation DiD exactly. Four-vs-eight-bit first-token agreement was 0.95; language consistency was 100%; batching-outcome agreement was 32/32.

2.4 Artifact 3: Base models and mechanistic induction (Experiment 2)

Workspace: `gen_art_experiment_2`.

Four checkpoints (gemma-3-12b-pt, GaMS3-12B, gemma-3-12b-it, GaMS3-12B-Instruct) were run in NF4 on an RTX 4090. Items were 2,889 RefusEU train+test pairs plus 400 alpaca harmless prompts. OpenRouter spend was \$4.74.

2.4.1 Alternate 2 (base-checkpoint geometry): killed

At the pre-registered layer ($L = 22$), the delta- d' -Slovene (GaMS-base minus pt) was -0.98 $[-1.13, -0.85]$: the GaMS base has *lower* Slovene separability than the pretrained ancestor at this layer, the opposite of the prediction. At each model’s own best layer, the sign reversed to $+2.85$. The item-level regression found that base geometry explained only 4.2% of the instruct-level refusal variance $[0.5\%, 12.2\%]$, with a z-score of -15 against the 50% mediation threshold.

Alternate 2 is dead. The base checkpoint does not set the instruct model’s language profile of refusal.

2.4.2 C4 (ancestor-direction induction): failure branch

The pretrained direction never reached a refusal rate of 0.50 in any model-language cell under the judge readout (peak 0.03 to 0.25; output was mostly incoherent). It performed no better than random directions. This triggers the pre-registered failure branch.

The positive control passed: each instruct model’s own difference-of-means direction induced refusal at low alpha values (alpha50 approximately 0.03 times the mean residual norm for Gemma-IT, 0.05 for GaMS3). The own-direction delta for the language-asymmetry comparison was -0.09 $[-0.28, 0.07]$ on the lexicon and $+0.24$ $[0.13, 0.38]$ on the judge, both inside the pre-registered equivalence bound of $\pm \ln(1.5) = 0.41$.

Table 5: SCORE-400 judge refusal rates (Experiment 2, Iteration 1, secondary).

Model	EN	SL
Gemma-IT	0.957	0.945
GaMS3-Instruct	0.967	0.917
DiD (log-odds)	-0.70 $[-1.45, 0.02]$	

2.5 Artifact 4: Persona-gate test (Experiment 3)

Workspace: `gen_art_experiment_3`.

Alternate 3 predicted that GaMS3’s Slovene-anchored persona gates Slovene refusal, producing a uniform surplus. The test ran GaMS3-12B-Instruct and Gemma-3-12B-IT in NF4 on SCORE-400 RefusEU pairs. Judge: gemini-2.5-flash, spend \$1.45.

2.5.1 Verdict: untestable

The manipulation check failed. Persona ablation removed own-naming, but a variance-matched random direction also removed English naming to a comparable degree. The pre-registered rule declared alternate 3 untestable.

Even if the manipulation had been clean, the signature would not survive. The judge-based triple-difference (TD*) was $+0.23$ $[-0.81, 1.25]$, with an MDE of 1.46.

Exploratory finding: ablating GaMS3’s own-name direction made GaMS3 identify as Qwen (English $0.15 \rightarrow 0.78$, Slovene $0.03 \rightarrow 0.60$). This is teacher-identity substitution, not persona removal.

Table 6: Refusal and identity rates under persona ablation (GaMS3, SCORE-400; Experiment 3, Iteration 1).

Condition	EN refusal	SL refusal	EN own-name	SL own-name
Original	0.925	0.675	0.45	0.90
Persona ablated (late)	0.925	0.653	0.00	0.00
Random control (late)	0.925	0.670	$0.23-0.30$	$0.23-0.30$

Alternate 3 is dead. The manipulation check fails, the premise (a Slovene surplus) is contradicted by the data, and the effect size is too small relative to the MDE.

2.6 Artifact 5: Heretic ablation and transfer curve (Experiment 4)

Workspace: `gen_art_experiment_4`.

English-objective ablation was run on both instruct checkpoints using Heretic (commit 3521f864) with identical configuration: NF4, empty system prompt, TPE sampler with seed 20260923. A driver script copied Heretic’s objective verbatim and logged bilingual metrics that never entered the objective.

2.6.1 English ablation reaches Slovene

Table 7: Refusal rates before and after Heretic ablation (lexicon, 400 pairs; Experiment 4, Iteration 1).

Condition	Gemma EN	Gemma SL	GaMS EN	GaMS SL
Original	0.965	0.932	0.940	0.830
Selected edit	0.242	0.223	0.133	0.098

Trial-level slopes of Slovene refusal on English refusal were positive for both models (Gemma 1.21 [0.75, 1.83]; GaMS3 0.79 [0.51, 1.14]). English-objective ablation removes Slovene refusal. This is a measurement of the de-censoring edit’s cross-lingual reach, relevant for evaluating the robustness of bilingual safety training.

2.6.2 C3 (frozen-core transfer): fails both ways

[**Correction (iter 4, MINOR #1)**: The exp4 summary gives $G_3 = -1.86$ $[-2.87, -1.13]$ on the matched first 17 trials, -1.76 on all trials, -0.98 on the lambda curve. The text below used -1.76 for the matched-17 value; the correct matched-17 value is -1.86 .]

The primary statistic G_3 was -1.86 $[-2.87, -1.13]$ on the matched first 17 trials (-1.76 on all trials). The main claim required $|G_3| < m = 0.675$; alternate 1 required $G_3 > m$. Neither holds: Slovene refusal is *more* exposed in GaMS3 than in Gemma at any matched English refusal level.

2.6.3 C5a (excess Slovene KL): reverses the prediction

The excess Slovene/English KL ratio was 0.298 [0.154, 0.562] for Gemma and 0.574 [0.409, 0.796] for GaMS3, both with confidence intervals entirely below 1. The real direction damages Slovene *less* than random directions.

2.6.4 C5b (bilingual re-selection): unpowered

Only one candidate trial per model met the re-selection criteria, and it was the same as the original pick. Gain was zero.

2.6.5 Alternate 4 (thin templated gate): strongest signal but does not survive

The prefill signature Sig was 3.63 [3.13, 4.36] in English and 1.54 [1.20, 1.92] in Slovene, both exceeding the m_2 threshold of 0.40. However, alternate 4 predicted symmetric language effects ($|\text{Sig}_{\text{SL}} - \text{Sig}_{\text{EN}}| < m$), and the observed asymmetry was 2.10, far exceeding the threshold. Alternate 4 was scored as not survived.

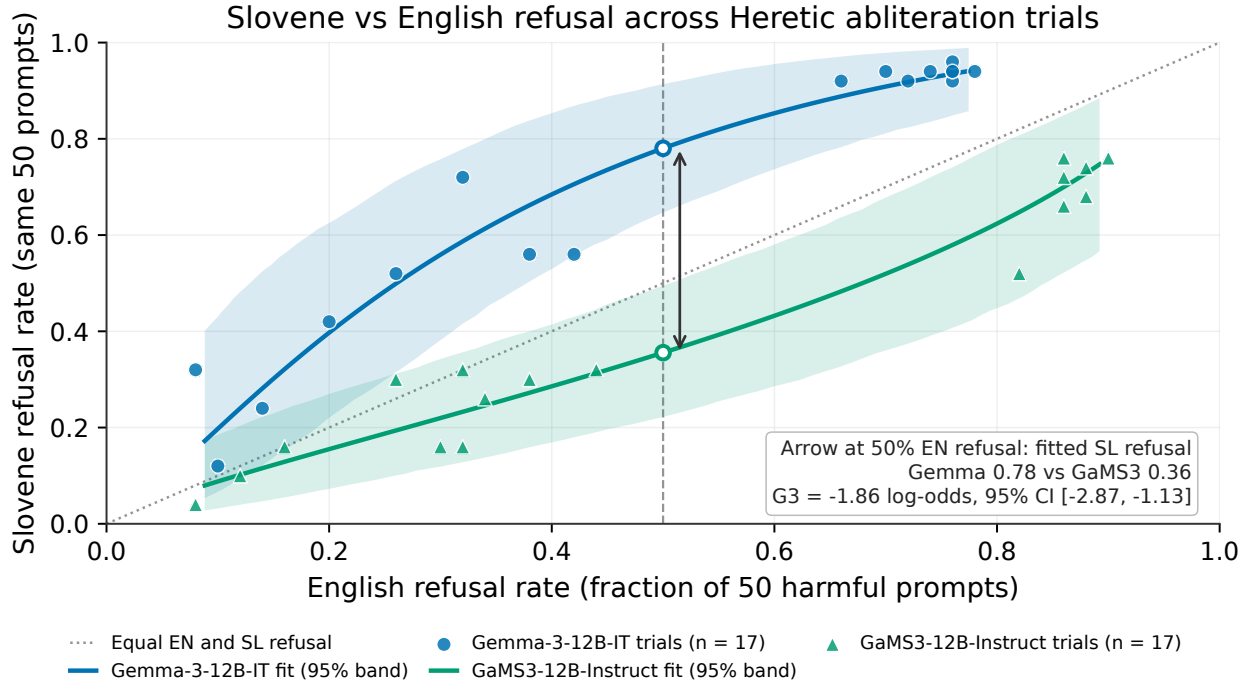


Figure 2: Slovene refusal rate against English refusal rate across English-objective Heretic ablation trials (Experiment 4, iteration-1 screen). Each marker is one trial’s edit, scored on 50 harmful RefusEU prompt pairs with the pre-registered refusal lexicon: blue circles are Gemma-3-12B-IT, green triangles are GaMS3-12B-Instruct, 17 trials each. Solid curves are each model’s logistic transfer fit. At 50% English refusal, the fits give Slovene refusal of 0.78 for Gemma and 0.36 for GaMS3. The arrow marks $G_3 = -1.86$ log-odds [95% CI $-2.87, -1.13$] (all 20 Gemma trials: $-1.76 [-2.73, -1.09]$). The lexicon readout was not validated against a judge. Later iterations did not confirm this lag under their pre-registered criteria, so it is a screen-level estimate.

Table 8: Prefill flip rates (Experiment 4, Iteration 1).

	Gemma EN	Gemma SL	GaMS EN	GaMS SL
Flip rate	0.031	0.558	0.559	0.855

2.6.6 Failed attempts and limits

The plan’s raw Arditi-style rank- k ablation destroyed Gemma (KL 20 to 53 nats even for a random basis) and was abandoned. Judge validation did not confirm the lexicon (kappa below 0.70 in all cells). GaMS3’s judge labels are incomplete: the OpenRouter key hit its daily limit.

All transfer-curve and Pareto numbers come from reduced-trial random search, not the planned 80-trial Optuna run.

2.7 Summary of dead ends and surviving claims after Iteration 1

Table 9: Status of pre-registered claims after iteration 1.

Claim	Status	Key evidence	Next step
Main: behaviour-level reach (C2)	INCONCLUSIVE	D uninformative; MDE 1.5–1.9	FINAL
Alt 1: supervised reach	INCONCLUSIVE	Same D ; zEN flips sign	Same
Alt 2: base geometry	DEAD	Delta- d' sign depends on layer	None
Alt 3: persona gate	UNTESTABLE DEAD	/ Manipulation check fails	None
Alt 4: thin gate	NOT SURVIVED	Prefill asymmetry too large	Descriptive
C1: identity calibration	PASS	$\text{DiD}_{\text{id}} = 3.22$	Carry forward
C3: frozen-core transfer	FAILS BOTH WAYS	$G_3 = -1.86$	Investigate
C4: ancestor induction	FAILURE BRANCH	pt direction fails	Report
C5a: excess SL KL	REVERSES	Ratio 0.30 / 0.57, CI below 1	Confirm
C5b: re-selection	UNPOWERED	$n = 1$ candidate	Full run

3 Iteration 2

3.1 Strategy

Iteration 2 addressed three weaknesses. First, the main claim remained inconclusive because the dose contrast was unpowered. Second, the Heretic transfer-curve finding rested on a lexicon readout that failed judge validation. Third, the teacher-inheritance pathway was flagged as a plausible alternative account.

The iteration ran five artifacts:

1. Experiment 5 (confirmation): the pre-registered FINAL evaluation on the reserved RefusEU evaluation split, using a local Qwen3-14B judge after the OpenRouter budget was exhausted.
2. Experiment 6 (refusal depth pipeline): code scaffold, no scored output.
3. Experiment 7 (teacher inheritance): a five-system screen comparing GaMS3, Gemma-IT, Qwen3-235B, Qwen3-14B, and Llama-3.3-70B.
4. Experiment 8 (Heretic re-test): a fresh paired ablation run on 200 item-matched probes with three dose curves.
5. Evaluation 1 (reconciliation): a \$0 harmonised re-scoring of all iteration-1 screens.

The OpenRouter shared key was exhausted for most of the iteration.

3.2 Artifact 6: Confirmation on the reserved FINAL splits (Experiment 5)

Workspace: `gen_art_experiment_5`.

Both models were run in NF4 quantization with greedy decoding (160 tokens, empty system turn) at seed 20260924. **Deviation D5:** the OpenRouter budget was exhausted mid-FINAL; the FINAL primary readout is a local Qwen3-14B judge validated against gemini on DEV (binary kappa 0.68 to 0.86).

3.2.1 Identity calibration: passes on FINAL

GaMS3 named itself “GaMS” in 90% of Slovene answers and 33% of English answers. GaMS3 credited Qwen or Alibaba in 51% of English identity responses. The FINAL identity DiD was 3.46 [2.55, 4.62].

Table 10: Identity results on FINAL (80 pairs; Experiment 5, Iteration 2).

Cell	Own-name rate (regex)
GaMS EN	0.325
GaMS SL	0.900
Gemma EN	0.563
Gemma SL	0.413

3.2.2 Refusal on RefusEU natural pairs

Table 11: FINAL refusal rates (Qwen3-14B judge, RefusEU natural pairs; Experiment 5, Iteration 2).

Model	EN	SL
GaMS3-Instruct	0.972	0.922
Gemma-IT	0.979	0.958

The overall DiD was -0.35 $[-0.81, 0.13]$, about -2.8 pp. The 95% CI includes zero. The dose contrast was -0.27 $[-1.31, 0.71]$, with MDE 1.36 exceeding $2m = 1.02$.

Table 12: DiD by judge family on the same 607 pairs (Experiment 5, Iteration 2).

Judge	DiD [95% CI]
Gemini-2.5-flash	-0.63 $[-1.14, -0.16]$
Qwen3-14B	$+0.03$ $[-0.75, 0.82]$

This judge dependence is the iteration’s most consequential finding: the size and sign of the refusal DiD are not robust to the choice of judge family.

3.2.3 HARD set: signal-detection analysis

The DiD in d' (discrimination) was -0.003 $[-0.33, 0.29]$: the two models lose the same amount of discrimination between languages. The DiD in criterion c was $+0.52$ $[0.38, 0.68]$, the most robust

Table 13: Signal-detection parameters on HARD (SL-MT vs EN-BT; Experiment 5, Iteration 2).

	d'	c (criterion)	Hit rate	False-alarm rate
Gemma EN	1.574	−0.440	0.890	0.364
Gemma SL	1.393	−0.870	0.941	0.569
GaMS EN	1.862	−0.327	0.896	0.273
GaMS SL	1.678	−0.235	0.859	0.273

result of the FINAL run. Gemma-IT shifts its criterion toward refusal on benign Slovene prompts (false-alarm rate rising from 0.364 to 0.569), while GaMS3 is language-invariant on criterion (false-alarm rates 0.273 in both languages).

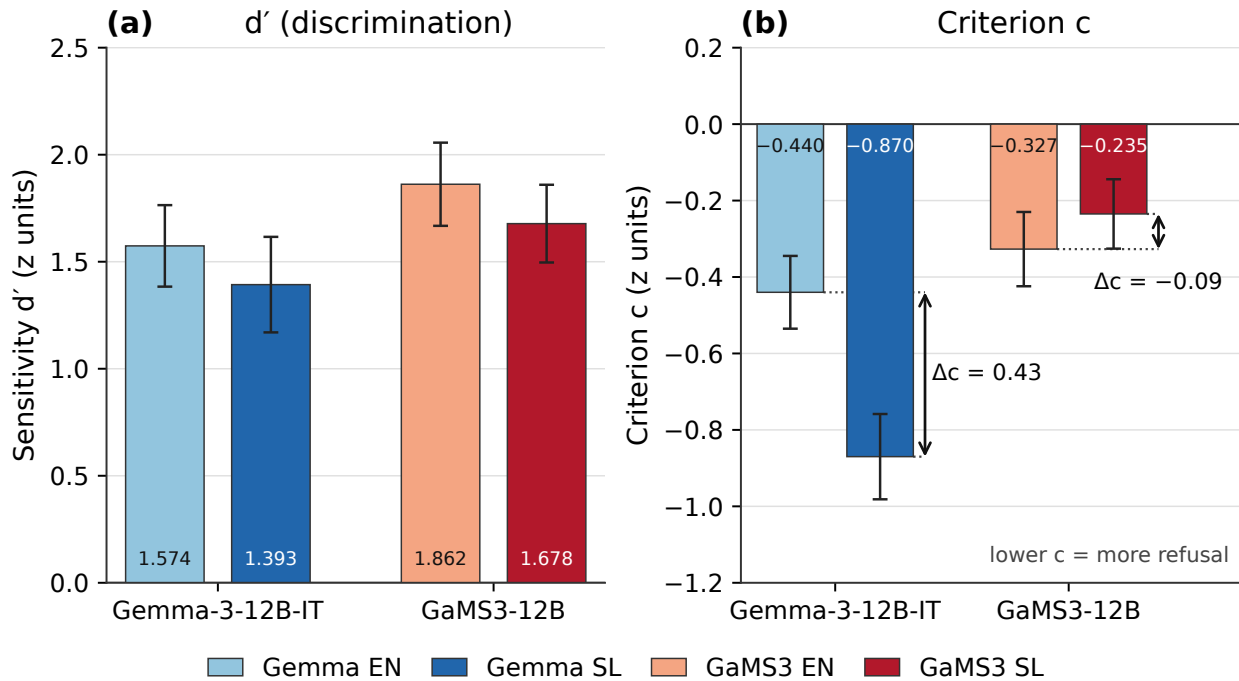


Figure 3: Signal-detection parameters for the unedited models on the HARD set (349 unsafe + 1,093 safe XSTest/OR-Bench items; refusals labelled by a local Qwen3-14B judge). Bars are Gemma-3-12B-IT (blue) and GaMS3-12B-Instruct (red), each in English (light) and Slovene (dark). (a) Sensitivity d' . (b) Criterion c (lower c = more refusal). The two models lose a similar amount of discrimination in Slovene (DiD in $d' = -0.003$ [−0.33, 0.29]). The criterion DiD is +0.52 [0.38, 0.68]: Gemma’s criterion moves toward refusal on Slovene items, while GaMS3’s false-alarm rate stays at 0.273 in both languages.

3.2.4 Refusal style

GaMS3’s Slovene refusals on RefuseEU are terse (median 16 tokens, vs 160 for Gemma) and formulaic: 61% open with “Oprostite, vendar vam pri tem” and 87% are covered by the top five openings. In English, 86% of GaMS3 refusals open with “I’m sorry, but I can’t.” Gemma’s refusals are longer (median 160 tokens), with more varied openings.

3.3 Artifact 7: Teacher-inheritance screen (Experiment 7)

Workspace: `gen_art_experiment_7`.

This exploratory screen tested whether GaMS3’s refusal decisions and wording inherit from its Qwen-family teacher. Five systems were run on byte-identical prompts.

Readout caveat. Headline numbers use a local Mistral-Small-24B judge, validated against gemini with binary kappa 0.67 to 0.86.

3.3.1 Decision inheritance: uninterpretable

The fingerprint-calibration gate failed. All pairwise kappas cluster at 0.51 to 0.60, dominated by item difficulty (retest ceiling 0.88).

3.3.2 Wording inheritance: supported

The SFT-source classifier (5-fold cross-validated accuracy 0.97) attributed 95.8% of GaMS3’s English refusal first sentences to the Qwen teacher family. In Slovene, GaMS3’s refusal wording matches the GaMS-27B translated SFT data (90.8% of first sentences found verbatim) rather than the live teacher’s output (1.5%).

Table 14: Refusal first-sentence origin (share found verbatim in SFT data; Experiment 7, Iteration 2).

Model	EN	SL
GaMS3-Instruct	0.761	0.908
Gemma-IT	0.000	0.030
Qwen3-235B	0.757	0.015
Qwen3-14B	0.092	0.007
Llama-3.3-70B	0.254	0.016

3.4 Artifact 8: Heretic transfer-curve re-test (Experiment 8)

Workspace: `gen_art_experiment_8`.

This artifact re-tested the iteration-1 finding using a design that removes the iter-1 confounds. It uses a fresh 200-item RefusEU-TRAIN probe, item-matched across three arms.

[Correction (iter 4, MUST-FIX #6a): The primary readout was the unvalidated lexicon. The Llama-3.1-8B judge labelled only 3,374 of 24,000 rows (kappa 0.52).]

Table 15: G_3 across dose curves (Experiment 8, Iteration 2).

Curve	G_3 [95% CI]	$m=0.675$	$m_{loc}=0.20$
B (29 paired)	−2.36 [−3.18, −1.62]	SL.OVER.EXP	SL.OVER.EXP
A1 (lambda)	−0.19 [−1.03, 0.78]	INCONCLUSIVE	INCONCLUSIVE
A2 (mean-proj)	excluded (non-specific)	n/a	n/a

The confirmatory curve confirms and strengthens the iter-1 finding (iter-1: intercept = −1.86 [−2.87, −1.13] on 17 trials). The lambda-scaling curve gives a much smaller and non-significant intercept (−0.19): the two curves disagree.

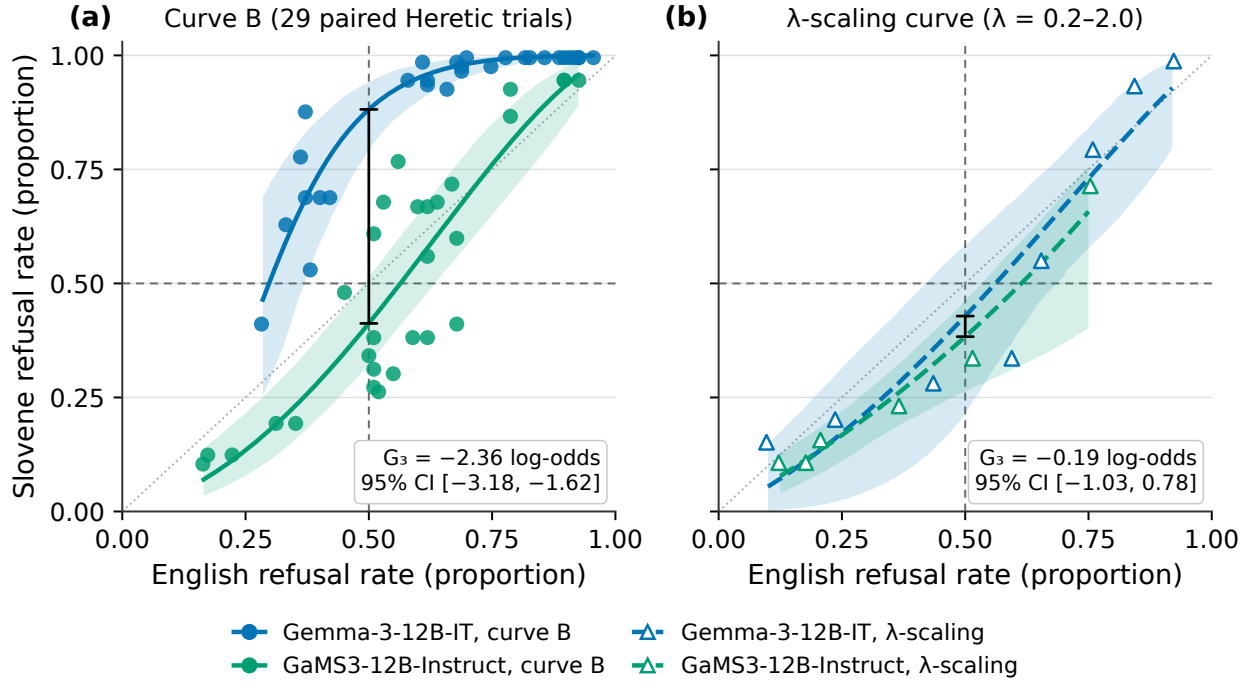


Figure 4: Slovene refusal rate as a function of English refusal rate after English-objective Heretic ablation (Experiment 8). (a) Confirmatory curve B: 29 paired fresh Heretic trials per model. (b) λ -scaling of the iteration-1 selected edit. Lines are per-model binomial GLM fits. On curve B, $G_3 = -2.36 [-3.18, -1.62]$. On the λ -scaling curve, $G_3 = -0.19 [-1.03, 0.78]$; the two dose curves disagree. The lexicon readout is unvalidated.

The excess Slovene/English KL ratio was 0.54 [0.34, 0.80] for Gemma and 0.93 [0.77, 1.15] for GaMS3, both at or below 1.

3.5 Artifact 9: Reconciliation of iteration-1 findings (Evaluation 1)

Workspace: `gen_art_evaluation_1`.

This \$0 evaluation harmonised all iteration-1 screens onto a single readout. 33 cells passed, 17 failed, and 22 were unvalidated.

Key finding 1: Prefill cells are unmeasurable. Against a 48-item blind adjudication, no readout reached kappa 0.60 on prefill continuations. The prefill-depth signature (alternate 4) and the iter-1 transfer-curve estimate are therefore uncitable.

Key finding 2: GaMS3 Slovene deflection. 34.6% of GaMS3’s Slovene prefill continuations paraphrased the harmful request back, a pattern absent in all other cells (0%).

Key finding 3: Meta-analysis. A generalised least-squares meta-analysis gave -0.50 $[-0.82, -0.21]$; $I^2 = 0.40$.

Table 16: Reconciliation verdict table (abbreviated; Evaluation 1, Iteration 2).

Claim	Iter-1 verdict	Harmonised verdict	Changed?
Main/Alt-1 (<i>D</i>)	INCONCLUSIVE	INCONCLUSIVE	N
Alt-2 (base geometry)	DEAD	DEAD	N
Alt-3 (persona gate)	UNTESTABLE	UNTESTABLE	N
Alt-4 (prefill depth)	NOT SURVIVED	UNCITABLE	Y
C1 (identity)	PASS (3.22)	PASS (3.22)	N
C3 (transfer curve)	FAILS BOTH (-1.86)	UNCITABLE (-0.44 prov.)	Y
C4 (ancestor induction)	FAILURE BRANCH	FAILURE BRANCH	N
C5a (excess SL KL)	REVERSES (< 1)	REVERSES (reproduced)	N
C5b (re-selection)	UNPOWERED	UNPOWERED	N

3.6 Artifact: Refusal-depth pipeline (Experiment 6)

Workspace: `gen_art_experiment_6`.

This artifact measured refusal depth across prefill lengths in both models (NF4, greedy, 160 tokens, gemini-2.5-flash judge with gpt-4.1 second family). The judged readout was **not validated**: against blind adjudication ($n = 50$), gemini kappa was 0.185 and all 19 errors ran in the direction that would manufacture a language gap. The primary result is therefore the judge-free representation readout.

H_depth verdict: FAIL. The pre-registered refusal-depth contrast did not pass.

Causal restoration (adding $a \cdot \text{SD} \cdot u_{\text{ref}}$ at L^* from the prefix end onward) was generated but **UNJUDGED** for both models; the causal claim is not established in this artifact.

4 Iteration 3

4.1 Strategy

Three open problems guided this iteration. First, the experiment 8 transfer-curve headline rested on the lexicon and a Llama-3.1-8B judge, neither validated against blind adjudication on edited rows. Second, two mechanistic hypotheses remained untested: whether the Slovene refusal that

Table 17: Label-free representation depth at L^* (all D300 items; retention 1 = refused-harmful DEV anchor, 0 = harmless; Experiment 6, Iteration 2).

Model	Cond	ret_ref SL-MT	ret_ref EN-BT	SL – EN-BT	95% CI	n
Gemma-IT	k0	1.022	1.015	0.007	[−0.001, 0.014]	300
Gemma-IT	P3	0.445	0.325	0.120	[0.113, 0.126]	300
Gemma-IT	P5	0.190	0.442	−0.252	[−0.256, −0.248]	300
Gemma-IT	P10	0.148	0.173	−0.025	[−0.028, −0.023]	300
Gemma-IT	P20	0.228	0.413	−0.184	[−0.192, −0.177]	300
Gemma-IT	Pfull	0.235	0.422	−0.187	[−0.196, −0.178]	300
Gemma-IT	N5	0.292	0.332	−0.040	[−0.042, −0.038]	300
Gemma-IT	Nfull	0.706	0.600	0.105	[0.098, 0.112]	300
GaMS3	k0	0.986	0.985	0.001	[−0.004, 0.006]	300
GaMS3	P3	0.674	0.414	0.259	[0.251, 0.268]	300
GaMS3	P5	0.288	0.695	−0.408	[−0.423, −0.393]	300
GaMS3	P10	0.249	0.413	−0.165	[−0.174, −0.155]	300
GaMS3	P20	0.550	0.417	0.133	[0.126, 0.139]	300
GaMS3	Pfull	0.519	0.507	0.012	[0.006, 0.018]	300
GaMS3	N5	0.530	0.784	−0.254	[−0.260, −0.247]	300
GaMS3	Nfull	0.426	0.426	0.001	[−0.002, 0.003]	300

Table 18: Judge sensitivity under each instrument, gemini vs local judge (Experiment 6, Iteration 2). Flagged: judged readout is NOT validated.

Readout	Model	J	DG	95% CI	DGN
gemini	Gemma-IT	471	−1.088	[−2.034, −0.407]	−0.858
gemini	GaMS3-Instruct	460	−0.618	[−1.046, −0.226]	−0.057
local	Gemma-IT	0	—	—	—
local	GaMS3-Instruct	16	—	—	—

outlasts English ablation is carried by a geometrically separate “Slovene caution direction” (the geometry hypothesis, M-a), or whether the refusal direction merely has greater induction potency in Slovene (the induction hypothesis, M-b). Third, no utility control had been run.

The iteration ran five artifacts: Evaluation 2, Experiment 9, Experiment 10, Experiment 11, and Experiment 12. The OpenRouter budget was exhausted throughout.

4.2 Artifact 10: Screen repair of experiment 8 (Evaluation 2)

Workspace: `gen.art.evaluation_2`.

This evaluation re-judged experiment 8’s saved responses (all 24,000 rows) with two fresh local judges: Qwen3-14B and Mistral-Small-24B.

[Correction (iter 4, MUST-FIX #6c): Qwen3-14B was selected because only it had full coverage (19,971 rows vs 2,614 for Mistral-Small-24B), not because of higher per-cell kappa. Calibration kappa was 0.36 overall (EN 0.25).]

The experiment 8 lexicon transfer-curve intercept of -2.36 was reproduced bit-for-bit from the saved response files. Under the validated judge, the integrated gap is -1.95 $[-3.75, -0.78]$ under REFUSE-only coding, but becomes INCONCLUSIVE under PARTIAL-as-refusal coding.

[Correction (iter 4, MUST-FIX #7): The tipping analysis was reported one-sidedly. The Gemma-SL false-REFUSE path is robust, but the GaMS-SL false-COMPLY path is not: $\varepsilon^* = 0.098$ to reach $-m$ (0.143 to reach 0), which requires only a 10% error rate.]

Table 19: Judge kappa against blind adjudication on edited rows (Evaluation 2, Iteration 3).

Instrument	Kappa (binary-R)
Gemini-2.5-flash	0.43
Qwen3-14B	0.26
Llama-3.1-8B	0.22
Lexicon	0.20

4.3 Artifact 11: Cross-lingual ablation lag confirmation (Experiment 9)

Workspace: `gen.art.experiment_9`.

Both models were run in 8-bit floating-point (FP8) quantization on a fresh 300-item probe. The primary judge was a DeBERTa classifier distilled from 11,767 archived gemini labels (calibration kappa 0.84).

[Correction (iter 4, MUST-FIX #6b): The DeBERTa was trained on 12,639 labels (9,884 train / 2,755 val), not 11,767.]

On the 9-point lambda curve, the transfer-curve intercept was -0.97 $[-1.56, -0.44]$, exceeding the pre-registered margin ($m = 0.675$). Under the second judge family (Llama-3.1-8B), it was -1.72 $[-2.30, -0.96]$.

The excess Slovene/English KL divergence ratio was 0.32 $[0.16, 0.62]$ for Gemma and 0.62 $[0.43, 0.87]$ for GaMS3, replicating.

4.4 Artifact 12: Mechanism test (Experiment 10)

Workspace: `gen.art.experiment_10`.

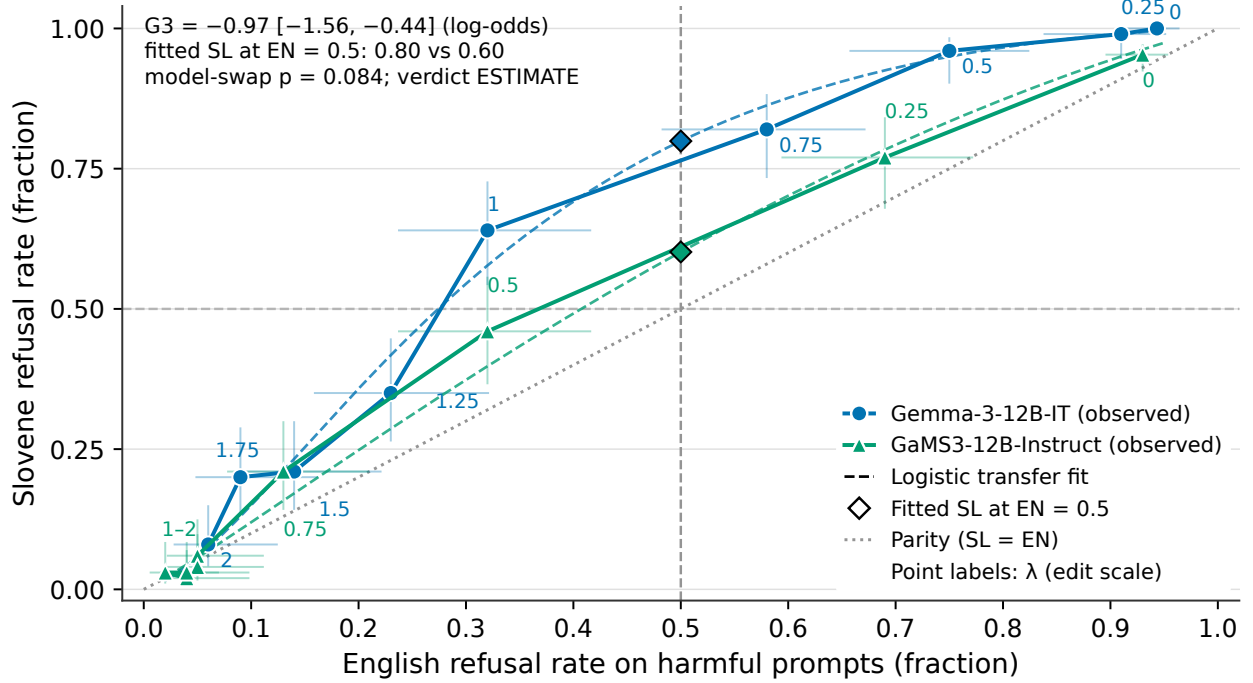


Figure 5: Slovene vs. English refusal along the Experiment 9 ablation-scaling curve. Each point is one λ step. Blue circles: Gemma-3-12B-IT; green triangles: GaMS3-12B-Instruct. $G_3 = -0.97 [-1.56, -0.44]$ (log-odds, 95% bootstrap CI) exceeds the pre-registered margin $m = 0.675$. However, the model-label permutation gives $p = 0.084$, and the pre-registered verdict is ESTIMATE.

Table 20: Unedited refusal rates on the 300-item fresh probe (DeBERTa judge; Experiment 9, Iteration 3).

Model	EN harmful	SL harmful	EN benign	SL benign
Gemma-IT	0.94	1.00	0.08	0.26
GaMS3-Instruct	0.93	0.95	0.04	0.18

This experiment tested the geometry and induction hypotheses. Items were 160 TEST harmful + 100 benign + 100 Dolly harmless items. The judge was a TF-IDF surrogate trained on 11,767 archived gemini labels (kappa 0.68 EN, 0.82 SL).

4.4.1 Geometry: EN and SL refusal directions are nearly collinear

Table 21: Geometry of refusal directions at L^* (Experiment 10, Iteration 3).

Model	$\cos(r_{\text{EN}}, r_{\text{SL}})$	$f(\text{SL energy in EN dir.})$	Ceiling cos	Perp share
Gemma-IT	0.991 [0.989, 0.993]	0.983 [0.978, 0.986]	0.999	0.13
GaMS3-Instruct	0.986 [0.984, 0.988]	0.973 [0.970, 0.975]	0.998	0.16

[**Correction (iter 4, MUST-FIX #9e)**: At own- L^* , the model difference in f is 0.010, negligible. At a common layer, the sign reverses ($f = -0.009$ $[-0.014, -0.005]$). The geometry result is layer-dependent.]

Geometry hypothesis at own- L^* : FALSE for both models.

4.4.2 Induction: the Slovene-perpendicular direction does not induce refusal

Adding the Slovene-perpendicular component ($u_{\text{SL}\perp}$) to harmless activations at L^* never induced refusal above 50%. The $u_{\text{SL}\perp}$ direction is inert as a refusal inducer. **Induction hypothesis verdict: NOT_SUPPORTED.**

4.4.3 Add-on neutralisation

[**Corrections (iter 4, MUST-FIX #9b-d)**: $u_{\text{SL}\perp}$ cut-vs-random at $s = 1.5$ gives 0.62 [0.38, 0.97] (excludes 0 but fails EN-invariance gate at -18 pp EN). GaMS $\text{Lag}(E_0) = -0.33$ $[-0.70, 0.07]$ fails the add-on precondition; GaMS add-on results are uninformative. u_{lang} failed the manipulation check in Gemma.]

Only ablating the full Slovene refusal direction (A_{SLfull}) closed the lag (-6.17 $[-6.50, -5.87]$), at the cost of 50% degenerate outputs and $19\times$ KL.

Both the geometry hypothesis and the induction hypothesis are REFUTED. The lag is a property of the shared refusal subspace, not a geometrically distinct Slovene reserve.

4.5 Artifact 13: FINAL confirmation on reserved split (Experiment 11)

Workspace: `gen_art_experiment_11`.

[**Corrections (iter 4, MUST-FIX #4)**: (a) Quantization was NF4, not FP8. (b) Per-model n differing (Gemma 200 vs GaMS 1101 EN / 652 SL). (c) An 8-step FINAL lambda curve was run. (d) Mistral second-family gate had kappa 0.26 on edited rows. (e) Permutation $p = 0.092$.]

The transfer-curve intercept on the 8-step FINAL lambda curve was -0.60 $[-1.31, +0.40]$, verdict ESTIMATE. Permutation $p = 0.092$.

4.5.1 Hungarian arm

On the 122-item Hungarian arm, neither language shows a significant model difference. The difference in lag change (GaMS3 minus Gemma) on Hungarian was -0.46 $[-2.21, 1.44]$, INCONCLUSIVE. On Slovene, $+0.05$ $[-1.49, 2.02]$, also null.

Table 22: McNemar rates on the reserved split (Qwen3-14B judge; Experiment 11, Iteration 3).

Model	EN orig	EN edit	SL orig	SL edit
Gemma-IT	0.98	0.645	0.99	0.68
GaMS3-Instruct	0.97	0.58	0.97	0.46

[**Correction (iter 4, MUST-FIX #8)**: Neither language shows an effect; “Hungarian behaves like Slovene” carries no information about the lag.]

4.5.2 Belebele comprehension: unchanged by the edit

Table 23: Belebele accuracy (200 items; Experiment 11, Iteration 3).

Model	Condition	EN	SL	HU
Gemma-IT	Original	0.930	0.885	0.880
Gemma-IT	Edit (lam=1)	0.930	0.885	0.880
GaMS3	Original	0.905	0.875	0.870
GaMS3	Edit (lam=1)	0.910	0.870	0.825

4.5.3 Mixed-effects model

[**Correction (iter 4, MUST-FIX #2)**: A positive `gams:sl:edit` = +1.06 means the edit shrinks GaMS3’s SL-EN margin LESS than Gemma’s. The model difference in SL-EN margin is largely baseline (`gams:sl` = -1.73 , $p = 3 \times 10^{-5}$). This is evidence against an edit-induced lag on FINAL, not for it.]

The `gams:sl` coefficient was -1.73 ($p = 3 \times 10^{-5}$; VB posterior -1.83), and the `gams:edit` main effect was non-significant (-0.18 , $p = 0.47$). On FINAL, the model difference in SL-vs-EN refusal is largely pre-edit, not edit-induced.

4.6 Artifact 14: Utility control (Experiment 12)

Workspace: `gen_art_experiment_12`.

This experiment measured whether the English-only ablation edit damages Slovene downstream skills more than English ones, using `lm-eval` on six zero-shot multiple-choice tasks with 300 paired English/Slovene items per task.

Table 24: Catastrophe gate (300-item full sets; Experiment 12, Iteration 3).

Model	Condition	EN loss (95% hi)	SL loss (95% hi)	Verdict
GaMS3	Iter-1 edit	0.018 (0.039)	-0.007 (0.021)	OK
GaMS3	Art-2 edit	0.020 (0.043)	0.009 (0.038)	OK
Gemma	Iter-1 edit	-0.008 (0.009)	-0.022 (0.024)	OK
Gemma	Art-2 edit	-0.016 (0.002)	-0.049 (-0.000)	OK

Table 25: Language asymmetry (Experiment 12, Iteration 3).

Model	Condition	A [95% CI]
GaMS3	Iter-1 edit	+0.025 [−0.009, 0.060]
GaMS3	Art-2 edit	+0.011 [−0.023, 0.050]
Gemma	Iter-1 edit	+0.014 [−0.036, 0.082]
Gemma	Art-2 edit	+0.033 [−0.020, 0.125]

One localised cost: GaMS3 lost 3.7 pp on English ARC-Challenge under the iteration-1 edit (McNemar Holm-corrected $p = 0.041$) and 2.7 pp under the artifact-2 edit. BoolQ showed replicated losses (−2.0/−2.3 pp).

Table 26: KL footprint (Experiment 12, Iteration 3). First-token and 32-token reported separately.

Model	Cond	1st SL/EN	1st EXCESS [CI]	KL EN 32t	KL SL 32t	32t SL/EN
GaMS3	Iter-1	0.52	0.47 [0.38, 0.60]	0.024	0.010	0.39
GaMS3	Art-2	0.54	0.49 [0.40, 0.62]	0.029	0.011	0.38
Gemma	Iter-1	0.72	0.63 [0.34, 1.17]	0.073	0.019	0.26
Gemma	Art-2	0.72	0.63 [0.34, 1.15]	0.076	0.021	0.28

[Correction (iter 4, MUST-FIX #10): Table mixed first-token and 32-token quantities. The exp11 Belebele contradicts exp12 (GaMS3 HU drops 4.5 pp in exp11 but is unchanged in exp12). The ARC-C cost omits replicated BoolQ losses.]

Multi-token excess KL ratio (RESULTS.md): GaMS3 iter-1 0.57 [0.51, 0.66]; Gemma iter-1 0.29 [0.24, 0.37].

4.7 Not executed or abandoned in iteration 3

Table 27: Dead ends and unexecuted pre-registered components, Iteration 3.

Component	Status	Reason
C-EXT (public ablated checkpoints)	Not generated	Shared cache reclaimed mid-run
Frozen-core RQ2 edit (80 trials, bf16)	Never ran	All edits are iter-1 fallback
exp10 Qwen3-14B judge	Failed gate	kappa 0.25 EN / 0.33 SL
exp9 GaMS random-direction control	Not scored	$n_{\text{random_steps}} = 0$
exp11 job J6 (GaMS lambda=1 remainder)	Not reached	Run ended before completion
exp12 GSM8K	Cut	Budget exhaustion

5 Iteration 4

5.1 Strategy

Three problems guided this iteration. First, the C-LAG sign was consistent but no artifact’s pre-registered verdict reached CONFIRM; the main blocker was the edited-row readout (best kappa 0.43). Second, the mechanism question shifted to testing M-OUT (output-language effect) vs M-IN (input-language effect). Third, sparse lambda coverage near EN 50% increased uncertainty.

Five artifact slots were planned; slot 2 failed:

1. Evaluation 3: paid-judge re-scoring with gemini-2.5-flash and gpt-4.1-mini, plus Rogan–Gladen and PPI corrections.
2. Experiment 13: **FAILED** (GPU lost).
3. Experiment 14: $2 \times 3 \times 3$ input-language $\times$ output-language mechanism test.
4. Experiment 15: fine-grained lambda ladder.
5. Research artifact: prior-art and fact check.

5.2 Artifact 15: Readout completion with paid judge (Evaluation 3)

Workspace: `gen_art_evaluation_3`.

The primary judge was gemini-2.5-flash. Blind adjudication used two Claude-family adjudicators (40-row overlap, inter-adjudicator kappa 0.69).

5.2.1 Readout gate: FAILS

Table 28: Readout gate (gemini-2.5-flash vs adjudication; Evaluation 3, Iteration 4).

Cell	Se [95% CI]	Sp [95% CI]	$\kappa(\text{gemini-gpt})$	Pass
GaMS-SL-edited	0.91 [0.74, 0.97]	0.73 [0.61, 0.83]	0.39	False
Gemma-SL-edited	0.99 [0.87, 1.00]	0.66 [0.43, 0.83]	0.72	False

5.2.2 Pooled cross-model lag

Table 29: Pooled lag estimates (REML + HKSJ, $k = 3$ item bodies; Evaluation 3, Iteration 4).

Readout	G_3 [95% CI]	$G_{3,\text{edit}}$ [95% CI]	IG [95% CI]	I^2
Raw primary (gemini)	−1.17 [−1.84, −0.50]	0.39 [−1.95, 2.74]	−1.15 [−1.31, −0.99]	0.22
RG (unstable)	−1.79 [−7.17, 3.58]	n/a	−2.02 [−5.40, 1.35]	0.00
PPI	−1.77 [−1.89, −1.65]	0.20 [−0.72, 1.11]	−2.00 [−3.19, −0.82]	0.00
TTJ	−1.10 [−4.67, 2.47]	0.33 [−6.52, 7.18]	−1.40 [−2.46, −0.33]	0.59

The edit-induced component is centred near zero in every readout. **R-BASE (baseline offset) is the best-supported account.**

Table 30: Per-curve G_3 decomposition (raw primary readout; Evaluation 3, Iteration 4).

Curve	G_3 [CI]	$G_{3,\text{orig}}$ [CI]	$G_{3,\text{edit}}$ [CI]	On support	Verdict
exp9 lambda	−1.12 [−1.70, −0.63]	−2.56 [−4.07, −1.32]	1.43 [0.29, 2.98]	False	ESTIMATE
exp11 FINAL	−1.00 [−1.40, −0.68]	−1.11 [−2.74, −0.21]	0.10 [−0.85, 1.71]	True	CONFIRM-LAG
exp8 A1	−1.54 [−2.11, −1.00]	−1.13 [−2.67, 0.04]	−0.41 [−1.65, 1.26]	False	CONFIRM-LAG
exp8 B	−1.20 [−1.77, −0.58]	−1.33 [−2.37, 0.78]	0.12 [−2.21, 1.47]	True	ESTIMATE
exp10 op pt	$G_{3,\text{op}} = -2.42 [-4.35, -1.53]$		−3.40 [−6.09, −1.30]	—	CONFIRM-LAG

5.3 Artifact 16: $2 \times 3 \times 3$ mechanism test (Experiment 14)

Workspace: `gen.art.experiment_14`.

This experiment tested M-OUT and M-IN by crossing input language (EN, SL, HU) with forced output language (EN, SL, HU) at three ablation doses. Three models were run: Gemma, GaMS3, and p-e-w/gemma-3-12b-it-heretic. Items: 200 harmful + 100 benign.

[**Correction (iter 5):** Tables 28–30 and this section’s prose were regenerated from experiment 14’s own results file.]

Table 31: Gemma-IT refusal rate per input→output cell (Experiment 14, Iteration 4).

Cell	zero	lo ($\lambda = 0.5$)	hi ($\lambda = 1.0$)	Lang. compliance
EN→EN	0.88	0.59	0.18	1.00
EN→SL	0.95	0.92	0.77	0.99
EN→HU	0.94	0.81	0.46	0.98
SL→EN	0.88	0.46	0.07	1.00
SL→SL	0.95	0.85	0.56	1.00
SL→HU	0.94	0.84	0.53	1.00
HU→EN	0.88	0.54	0.10	1.00
HU→SL	0.95	0.93	0.83	0.96
HU→HU	0.94	0.80	0.55	1.00

Table 32: GaMS3-Instruct refusal rate per input→output cell (Experiment 14, Iteration 4). Cells below 0.90 compliance marked †.

Cell	zero	lo ($\lambda = 0.5$)	hi ($\lambda = 0.625$)	Lang. compliance
EN→EN	0.82	0.30	0.21	1.00
EN→SL	0.77	0.21	0.14	0.98
EN→HU	0.80	0.34	0.23	0.97
SL→EN	0.87	0.25	0.16	0.00†
SL→SL	0.87	0.20	0.15	1.00
SL→HU	0.82	0.27	0.17	0.56†
HU→EN	0.86	0.29	0.19	0.14†
HU→SL	0.92	0.23	0.12	0.86†
HU→HU	0.91	0.41	0.28	1.00

GaMS3 did not follow the output-language instruction off the English-input row. Its SL→EN compliance is 0.00. Only the English-input row is a valid output-language manipulation for GaMS3.

Cross-model contrasts: $dOUT_{SL} = -1.58 [-2.22, -0.66]$ (MDE 1.07); $dIN_{SL} = -0.26 [-1.13, 0.57]$ (MDE 1.19).

Verdict: ESTIMATE (screen). On the raw readout, edited Gemma’s residual refusal follows the output language. The call is not robust across readouts, and GaMS3 failed the output-language manipulation off the English-input row.

5.4 Artifact 17: Fine-grained lambda ladder (Experiment 15)

Workspace: `gen.art.experiment_15`.

Table 33: p-e-w/gemma-3-12b-it-heretic (public checkpoint, single external dose; Experiment 14, Iteration 4).

Cell	ext	Lang. compliance
EN→EN	0.05	1.00
EN→SL	0.15	0.92
EN→HU	0.02	0.96
SL→EN	0.01	0.99
SL→SL	0.21	1.00
SL→HU	0.07	0.99
HU→EN	0.04	0.94
HU→SL	0.14	0.68†
HU→HU	0.08	1.00

Table 34: Mechanism contrasts by readout (logit, 95% CI; Experiment 14, Iteration 4). Slovene contrasts only.

Readout	Model	OUT _{SL}	IN _{SL}	INT _{SL}	OUT−IN (SL)
raw_R	Gemma	1.14 [0.60, 1.56]	−0.60 [−1.02, −0.18]	−0.02	1.73 [1.03, 2.34]
raw_R	GaMS3	−0.44 [−0.98, 0.27]	−0.86 [−1.58, −0.13]	−0.66	0.41 [−0.55, 1.59]

Table 35: Lag at matched EN→EN refusal of 50% (logit, raw primary readout; Experiment 14, Iteration 4). GaMS3 values extrapolated.

Cell	Gemma a	Gemma L^*	GaMS3 a	GaMS3 L^*	G_3	$G_{3,\text{edit}}$
EN→EN	0.00	0.00	0.00	0.00	0.00	0.00
EN→SL	2.17	1.15	−0.45	−0.11	−2.63	−1.26
EN→HU	1.12	0.39	0.28	0.47	−0.85	0.09
SL→EN	−0.63	−0.59	−0.18	−0.53	0.45	0.06
SL→SL	1.46	0.54	−0.96	−1.30	−2.42	−1.84
SL→HU	1.34	0.52	−0.15	−0.15	−1.49	−0.67
HU→EN	−0.32	−0.32	0.08	−0.22	0.40	0.09
HU→SL	2.37	1.34	−0.00	−0.81	−2.37	−2.15
HU→HU	1.14	0.41	0.59	−0.10	−0.56	−0.51

Table 36: Signal-detection DiD (Gemma minus GaMS3) at dose zero and λ_{hi} (Experiment 14, Iteration 4).

Dose	Contrast	d' DiD	c DiD
zero	SL input	−0.02	0.20
zero	SL output	−0.13	−0.76 [−1.06, −0.52]
zero	SL both	−0.41	−0.47 [−0.78, −0.20]
hi	SL input	0.27	0.48 [0.07, 0.75]
hi	SL output	0.72	−1.56 [−1.97, −1.29]
hi	SL both	0.18	−1.16 [−1.72, −0.81]

This experiment ran a fine-grained lambda ladder – 13 steps for Gemma ($\lambda = 0\text{--}2.0$) and 15 steps for GaMS ($\lambda = 0\text{--}1.538$) – on a fresh 300-item RefusEU-TRAIN body.

Table 37: Headline G_3 decomposition (R coding, raw readout; Experiment 15, Iteration 4).

Readout	G_3 [95% CI]	$G_{3,\text{orig}}$ [95% CI]	$G_{3,\text{edit}}$ [95% CI]
RAW	−0.70 [−1.01, −0.42]	−1.17 [−2.83, 0.22]	0.47 [−0.96, 2.14]
RG	−0.01 [−4.36, 11.27]	−1.76 [−5.45, 1.96]	1.75 [−4.58, 13.18]
PPI	−0.75 [−2.43, 0.99]	−1.52 [−6.46, 2.82]	0.76 [−3.78, 6.27]
TTJ	−0.79 [−1.09, −0.48]	−1.37 [−2.29, 0.37]	0.58 [−1.18, 1.52]

$G_{3,R} = -0.70$ [−1.01, −0.42] with MDE 0.42, the most powerful estimate in the entire study. It survives under TTJ (−0.79 [−1.09, −0.48]). The edit-induced component spans zero: R-BASE is supported.

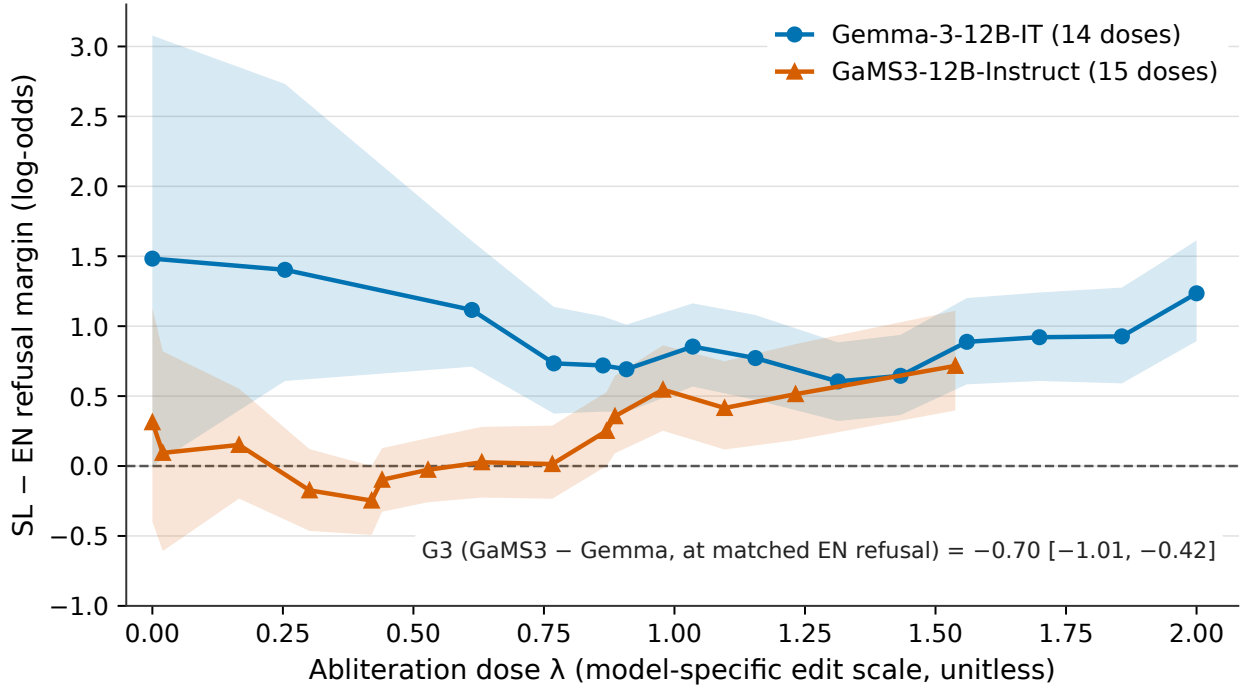


Figure 6: Slovene-minus-English refusal margin across the fine-grained ablation-dose ladder of Experiment 15, on 300 harmful RefusEU-TRAIN items per dose and language. Blue circles: Gemma (14 doses). Vermilion triangles: GaMS3 (15 doses). Gemma’s margin is positive at every dose (0.61–1.48 log-odds). GaMS3’s margin starts at 0.31, hovers near zero in the mid-range, and rises to 0.72 at $\lambda = 1.538$. The inset reports $G_3 = -0.70$ [−1.01, −0.42] (MDE 0.42). Judge specificity in the edited Slovene cells is only 0.55 (Gemma) and 0.62 (GaMS3).

5.5 Artifact 18: Prior-art and fact check (Research)

Workspace: `gen_art_research_1`.

No LLM spend; web greps plus one local overlap script. No published paper crosses prompt language with response language for refusal measurement as of 2026-09-24. The M-IN/M-OUT

design is novel (medium confidence). Nearest neighbours: Deng et al. 2023, Upadhyaya and Sikdar [11]. Heretic @3521f86 is a 2.0.0.dev0 snapshot (2026-09-05). Hungarian is not in the declared GaMS3 CPT/SFT mix.

The two figure markers [FIGURE:fig_mechanism_2x3] and [FIGURE:fig_rincap_composition] in the report text do not have corresponding pre-rendered figures; the experiment 14 and evaluation 4 data are presented in the tables above instead.

5.6 Not executed or abandoned in iteration 4

Table 38: Iteration-4 components not executed.

Component	Status	Reason
exp13 new Heretic edit and DEV lambda grid	Not executed	GPU lost
exp13 held-out confirmation curve (B-300)	Not executed	GPU lost
exp13 random-direction controls	Not executed	GPU lost
RQ2 natural RefusEU FINAL under edit	Not executed	GPU lost
exp13 substitute paid re-readout (19,195 rows)	Frozen, unlabelled	OpenRouter daily limit
exp13 judge tier gate	Ran; both tiers FAIL	Sp 0.27–0.55
exp14 harmful-content (ASR) readout	One cell only	Not completed

6 Iteration 5

6.1 Strategy

Three goals: (i) confirm C-OUT on a fresh body with PPI correction (experiment 16); (ii) run the deferred RefusEU FINAL evaluation with the full 1,300-item pool (experiment 17); (iii) test R-INCAP – whether Gemma’s Slovene reserve is real safety or production failure (evaluation 4). Two support artifacts: a paper-number audit (evaluation 5) and a novelty update (research 2).

C-LAG is now closed in favour of C-OUT. The novelty verdict is downgraded from NOVEL to PARTIALLY OCCUPIED after Addagada [34] and Nguyen et al. [35].

6.2 Artifact 19: C-OUT confirmation (Experiment 16)

Workspace: `gen_art_experiment_16`.

This experiment tested whether C-OUT replicates on a fresh 294-item body (146 StrongREJECT, 109 HarmBench, 39 JBB) with a new Heretic edit. The primary estimator was PPI (switched from Rogan–Gladden after the Youden-J diagnostic flagged instability).

Headline: $\text{OUT}_{\text{Gemma}}(\text{PPI}) = +2.43 [+0.89, +5.38]$ log-odds. Cross-model contrast $\text{dOUT} = -2.31 [-5.34, -0.68]$. GaMS3 shows no output-language effect: $\text{OUT}_{\text{GaMS3}} = +0.12 [-0.47, +0.72]$. The input-language effect in Gemma is negative: $\text{IN}_{\text{Gemma}} = -0.68 [-1.23, -0.36]$. All confirmation clauses pass but the verdict is capped at ESTIMATE because the harmful-content adjudication kappa is 0.12.

6.3 Artifact 20: RefusEU FINAL (Experiment 17)

Workspace: `gen_art_experiment_17`.

This experiment measured what one English-only de-censoring edit achieves on RefusEU’s own 1,300 prompts per language. Judge: PolyGuard refusal (judge-substitution deviation after OpenRouter budget exhaustion).

Table 39: Primary-judge refusal rates on harmful items, selected cells (Experiment 16, Iteration 5).

Model	Cell	Dose	Refusal
Gemma	EN→EN	hi	0.330
Gemma	EN→SL	lo	0.884
Gemma	EN→HU	lo	0.731
Gemma	SL→EN	lo	0.344
Gemma	SF_EN→EN	lo	0.510
Gemma	SF_SL>SL	lo	0.827
Gemma	EN→EN	zero	0.918
Gemma	EN→SL	zero	0.969
GaMS3	EN→EN	hi	0.490
GaMS3	EN→SL	hi	0.449
GaMS3	SF_EN→EN	hi	0.527
GaMS3	SF_SL>SL	hi	0.616
GaMS3	EN→EN	zero	0.820
GaMS3	EN→SL	zero	0.786

Table 40: Signal-detection DiD at dose zero and dose* (Experiment 16, Iteration 5).

Dose	Contrast	d' DiD	c DiD
zero	SL output	−0.37 [−0.81, 0.07]	−0.65 [−0.90, −0.46]
lo	SL output	+0.09 [−0.62, 0.57]	−1.07 [−1.42, −0.85]
zero	SL output (GaMS3 only)		+0.10 [−0.12, 0.35]
lo	SL output (GaMS3 only)		−0.05 [−0.42, 0.38]

Table 41: Judge error in decisive cells (LLM adjudication; Experiment 16, Iteration 5).

Cell	Judge vs reference	Se	Sp	n
Gemma EN→EN	gemini_U vs adj_U	0.93	0.63	99
Gemma EN→SL	gemini_U vs adj_U	1.00	0.94	99
GaMS3 EN→EN	gemini_U vs adj_U	0.92	0.84	100
GaMS3 EN→SL	gemini_U vs adj_U	1.00	0.82	99
Gemma EN→EN	J1 vs adj_explicit	1.00	0.46	100
Gemma EN→SL	J1 vs adj_explicit	1.00	0.35	100

Table 42: Paired edit effects on RefuseEU FINAL (Experiment 17, Iteration 5). All four cells measured as ROBUST.

Model	Lang	ASR orig	ASR edit	Delta [95% CI]	Verdict
Gemma	EN	0.045	0.897	0.852 [0.831, 0.870]	ROBUST
Gemma	SL	0.026	0.683	0.657 [0.629, 0.683]	ROBUST
GaMS3	EN	0.033	0.970	0.937 [0.923, 0.949]	ROBUST
GaMS3	SL	0.068	0.915	0.847 [0.826, 0.867]	ROBUST

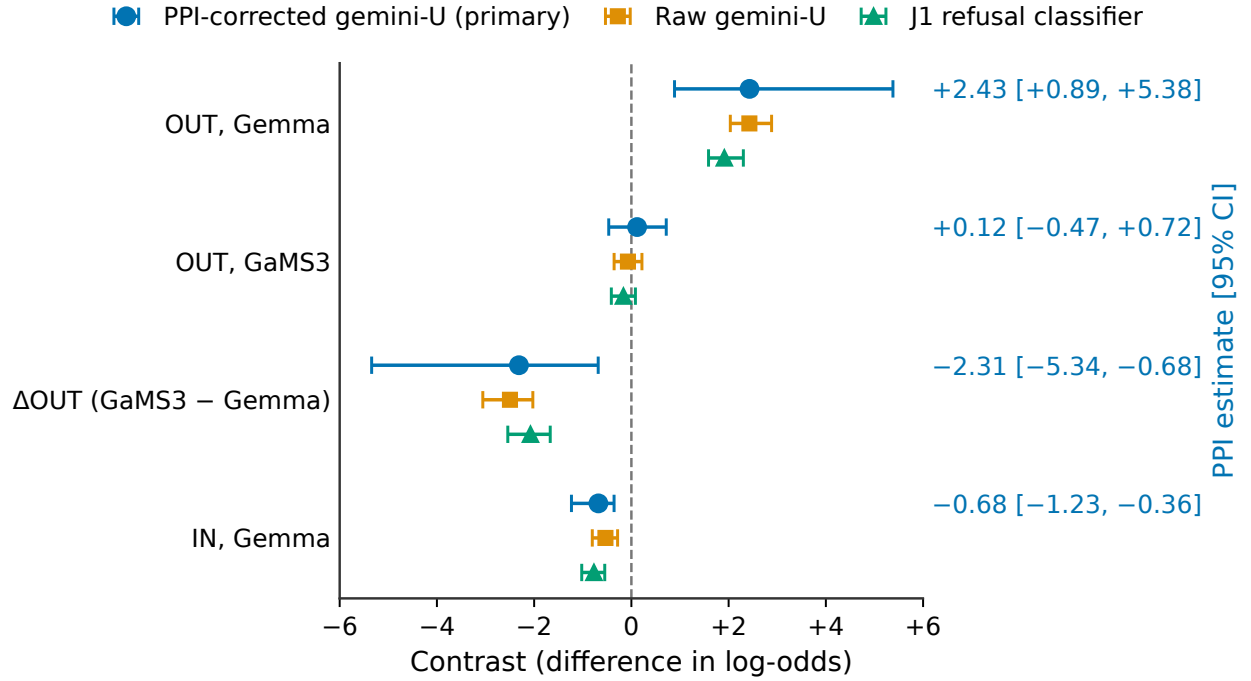


Figure 7: Output-language contrasts on the held-out 294-item confirmation set (Experiment 16), in log-odds. Gemma gives safer replies when it answers in Slovene (OUT = +2.43 [+0.89, +5.38]). GaMS3 shows no output-language effect (OUT = +0.12 [-0.47, +0.72]). The cross-model difference excludes zero (Δ OUT = -2.31 [-5.34, -0.68]). A Slovene input slightly lowers Gemma’s safe-reply rate (IN = -0.68 [-1.23, -0.36]). The verdict is capped at ESTIMATE ($\kappa_U = 0.12 < 0.6$). Adjudication is by LLMs, not humans.

Corrected ASR (Rogan–Gladen): Gemma EN 0.863 [0.811, 0.892]; Gemma SL 0.578 [0.473, 0.639]; GaMS3 EN 0.967 [0.927, 0.976]; GaMS3 SL 0.884 [0.826, 0.912].

The EN-vs-SL within-model difference is **not interpretable as a language effect** because the prompt sets differ (LaBSE 0.55). The ranking-vs-thresholding analysis shows the guards’ AUROC is high (0.887–1.000) but their decision thresholds are miscalibrated for Slovene.

6.4 Artifact 21: R-INCAP safety evaluation (Evaluation 4)

Workspace: `gen.art.evaluation.4`.

This artifact tested whether Gemma’s Slovene-output reserve is real safety or production failure. Harmful content was measured by StrongREJECT (original gemini rubric, translated rubric, and local fine-tuned).

At high dose: $OUT_R = 2.67$ [2.29, 3.12] and $OUT_U = 2.61$ [2.20, 3.07]. Both positive and exclude zero. Gemma refuses more in Slovene AND produces less harmful content in Slovene. **R-INCAP = HARMFUL CONTENT SEPARATES (reserve = safety).**

Table 43: Four-class composition at high dose, decisive cells (Evaluation 4, Iteration 5).

Cell	n	EXPLICIT	DEFLECT	COMPLY	PF index	U_{adj}
Gemma EN→SL hi	50	0.54	0.26	0.20	0.57	0.14
Gemma EN→EN hi	48	0.00	0.10	0.90	0.10	0.75
GaMS3 EN→SL hi	49	0.00	0.08	0.90	0.08	0.90
GaMS3 EN→EN hi	50	0.00	0.12	0.88	0.12	0.84

The production-failure index difference in Gemma is 0.46 [0.23, 0.65]; in GaMS3 it is -0.04 $[-0.17, 0.09]$. Half of Gemma’s Slovene non-compliance is deflection, consistent with Shen et al.’s relevance curse [27] and Addagada’s output-language effect [34].

6.5 Artifact 22: Paper number audit (Evaluation 5)

Workspace: `gen.art.evaluation.5`.

R-BASE ceiling sensitivity: Experiment 15’s baseline component (-1.17) is a CEILING-ARTEFACT (fragility index 1). Only the experiment 9 body has a ROBUST-OFFSET (fragility index 9).

Decomposition: Across $107 \text{ body} \times \text{readout}$ rows: 15 are edit-induced, 0 baseline, 64 under-determined. Parallel-curve geometry holds ($b_{\text{Gemma}} = 0.94$ [0.75, 1.17], $b_{\text{GaMS3}} = 0.80$ [0.63, 0.95]).

Criterion shift: 8 of 9 compliance-valid rows have a CI below 0.

Ledger: 1,176 numeric tokens plus 22 pointer claims: 332 SURVIVE, 844 UNTRACEABLE, 2 REVERSE, 1 SHRINKS.

6.6 Artifact 23: Novelty and fact check (Research 2)

Workspace: `gen.art.research.2`.

Main-claim novelty verdict: PARTIALLY OCCUPIED. Two new preprints:

- Addagada [34] (arXiv 2609.08373, 8 Sep 2026): forces the output language of an English structural jailbreak and scores with a StrongREJECT-style rubric. Non-English output lowers attack success in 11 of 12 cells.

- Nguyen et al. [35] (arXiv 2608.26186, 22 Aug 2026): fully crosses prompt and response language (EN/NO) on benign questions.

The narrowed design (crossing both factors with safety endpoints on a sibling pair under partial-dose ablation) remains unoccupied as of 2026-09-25 (medium-low confidence).

6.7 Not executed or abandoned in iteration 5

Table 44: Iteration-5 components not executed.

Component	Status	Reason
exp16 partial-dose points	Not executed	\$4.00 budget cap
exp16 Gemma high dose (partial)	Partial only	Deadline cut
exp16 huihui/mlabonne external checkpoints	Not run	Time
exp16 gpt-4.1-mini second adjudicator	Not run	OpenRouter budget
exp17 primary gemini judge	Replaced by PolyGuard	Budget exhausted
exp17 gemini StrongREJECT rubric	Replaced by fine-tuned	Budget exhausted
eval4 sonnet-4.5 on low dose	Not run	Budget
eval4 human audit of adjudication	Requested, not performed	No reviewer
Frozen-core Heretic (bf16, 80 trials)	Still not executed	Owed since iter 1
Natural RefusEU reserved-split under edit	Still not executed	Owed since iter 1

7 What We Have Learned

Five iterations screened the main claim and four alternates. Two alternates are dead (base-checkpoint geometry and persona gating). A third (prefill-depth signature) is uncitable. The geometry of refusal is layer-dependent: nearly collinear at each model’s best layer ($\cos = 0.986\text{--}0.991$) but divergent at a common layer. The perpendicular component does not induce refusal. The teacher-inheritance screen found that GaMS3’s refusal *decisions* do not detectably inherit from the Qwen teacher, while its refusal *wording* inherits almost entirely (76% of English and 91% of Slovene first sentences verbatim from the SFT data).

The primary contest (behaviour-level reach vs supervised reach) remains unresolved: the dose contrast is underpowered across all iterations. The overall GaMS3 Slovene deficit is judge-dependent; the meta-analytic estimate is -0.50 $[-0.82, -0.21]$. The criterion-shift finding replicates: 8 of 9 compliance-valid rows have CIs below zero (Gemma’s criterion moves toward refusing when it replies in Slovene; GaMS3’s does not). The shift is specific to the output language, not the input language, and is consistent with Aziz et al.’s calibration-not-representation account [7].

The cross-lingual ablation lag is better characterised as an output-language effect (C-OUT) than as a generic cross-lingual lag. The C-OUT finding replicates: $\text{OUT}_{\text{Gemma}}(\text{PPI}) = +2.43$ $[+0.89, +5.38]$, $\text{dOUT} = -2.31$ $[-5.34, -0.68]$. Gemma’s refusal follows the output language; GaMS3’s does not. The input-language effect is negative in Gemma (-0.68 $[-1.23, -0.36]$). All confirmation clauses pass but the verdict is capped at ESTIMATE (harmful-content $\kappa = 0.12$, anchor WEAK).

R-BASE remains the best-supported account for the overall lag. The edit-induced component spans zero (experiment 15: $+0.47$ $[-0.96, 2.14]$). The pre-edit baseline is negative everywhere. The curves are parallel. The R-BASE baseline is a ceiling artefact in most item bodies (fragility index 1 in experiment 15); only the experiment 9 body has a robust offset (fragility index 9).

The reserve is safety, not production failure. At both doses, harmful content separates (OUT_U original rubric = 2.61 [2.20, 3.07] at high dose). The four-class decomposition shows production-failure index difference 0.46 [0.23, 0.65] for Gemma, consistent with the relevance curse [27, 34].

On RefusEU FINAL prompts (experiment 17, $n = 1,300$ per cell), the edit is effective in all four cells (ROBUST). The edit widens the Slovene reserve (edit-induced output-language contrast = +2.43 [+0.92, +5.38]). English-only ablation does not close the Slovene-English refusal gap; it amplifies it.

The utility control (experiment 12) stands: the edit does not damage Slovene downstream skills more than English ones. The identity calibration arm continues to replicate (FINAL identity DiD = 3.46 [2.55, 4.62]).

The novelty verdict is PARTIALLY OCCUPIED after Addagada [34] and Nguyen et al. [35]. The narrowed design remains unoccupied.

Open items. The readout gate has never passed. The frozen-core Heretic procedure (bf16, 80 trials, Heretic’s own selection rule) has never been run. No artifact has scored natural RefusEU reserved-split prompts under the edit. All adjudication is author-LLM, not human.

8 Run Bookkeeping

Table 45: Run trajectory: one row per iteration with its move, number of artifacts, and cumulative judge spend. Spend excludes compute-rental time.

Iter	Move	Artifacts	Done?	Spend
1	Screen all pre-registered claims	5 (dataset, exp 1–4)	Yes	\$16
2	FINAL + teacher + reconciliation	5 (exp 5–8, eval 1)	Yes	\$21
3	Mechanism + utility + repair	5 (eval 2, exp 9–12)	Yes	\$21
4	Paid readout + 2×3×3 + ladder	4/5 (eval 3, exp 14–15, res 1)	Partial	\$25
5	C-OUT + FINAL + R-INCAP	5 (exp 16–17, eval 4–5, res 2)	Yes	\$29

References

1. A. Arditi et al. “Refusal in Language Models Is Mediated by a Single Direction.” NeurIPS 2024.
2. X. Wang et al. “Refusal Direction is Universal Across Safety-Aligned Languages.” NeurIPS 2025.
3. H. Du et al. “How Post-Training Reshapes LLMs.” arXiv 2504.02904, 2025.
4. U. Shaham et al. “Multilingual Instruction Tuning With Just a Pinch of Multilinguality.” Findings ACL 2024.
5. F. Joad et al. “There Is More to Refusal in Large Language Models than a Single Direction.” arXiv 2602.02132, 2026.
6. V. Zhong, Q. Li. “Refusal Lives Downstream of Persona in Chat Models.” ICML 2026 MI Workshop.
7. R. Aziz et al. “Low-Resource Safety Failures Are Action Failures, Not Representation Failures.” arXiv 2606.01196, 2026.
8. A. Krasnodebska et al. “Multilingual Refusal Alignment for Safer Large Language Models.” Findings ACL 2026.
9. W. Hawkins et al. “The Heterogeneous Safety Impacts of Benign Multilingual Fine-Tuning.” arXiv 2606.28843, 2026.

10. X. Li et al. “Preference Tuning For Toxicity Mitigation Generalizes Across Languages.” EMNLP 2024.
11. A. Upadhyaya, S. Sikdar. “When Safety Speaks a Language.” 2026.
12. S.-Y. Miao et al. “Who Bridges Safety?” 2026.
13. A. Oppong et al. “The Illusion of Cross-Lingual Safety in Low-Resource Languages.” 2026.
14. C. Kissane et al. “Base LLMs Refuse Too.” Alignment Forum, 2024.
15. L. Chua et al. “Crosslingual Capabilities and Knowledge Barriers in Multilingual Large Language Models.” arXiv 2406.16135, 2024.
16. X. He et al. “TUBA: Cross-Lingual Transferability of Backdoor Attacks.” Findings ACL 2025.
17. Y. Wu et al. “Edit Once, Update Everywhere.” Findings ACL 2025.
18. G. Messenger. “Detecting Safety Training Modification in Language Models via Activation Analysis.” IEEE Access, 2026.
19. A. Labunets. “Refusal geometry reflects refusal training.” 2026.
20. D. Vres et al. “Building a Strong Instruction Language Model for a Less-Resourced Language.” arXiv 2603.01691, 2026.
21. R. Young. “Comparative Analysis of LLM Abliteration Methods.” arXiv 2512.13655, 2025.
22. A. Fafula. “Abliteration Is Not a Scalpel.” arXiv 2607.17427, 2026.
23. N. Truong. “Abliteration Mitigation via Refusal Aliases.” arXiv 2608.18093, 2026.
24. P. E. Weidmann. “Heretic: Fully Automatic Censorship Removal for Language Models.” GitHub, 2025.
25. C. Lee et al. “How to Correctly Report LLM-as-a-Judge Evaluations.” ICML 2026. arXiv 2511.21140.
26. A. N. Angelopoulos et al. “PPI++: Efficient Prediction-Powered Inference.” arXiv 2311.01453, 2023.
27. L. Shen et al. “The Language Barrier: Dissecting Safety Challenges of LLMs in Multilingual Contexts.” Findings ACL 2024.
28. K. Y. Dahir. “SomaliBench Eval.” arXiv 2605.25420, 2026.
29. M. Zhang et al. “Why Do Safety Guardrails Degrade Across Languages?” arXiv 2605.17173, 2026.
30. C. Yoon et al. “Who Pays More for Safety?” arXiv 2608.22490, 2026.
31. P. Balani, S. Panda. “Hard Negatives Reveal What Easy Negatives Hide.” arXiv 2609.27758, 2026.
32. R. P. Kompella, A. Mahajan. “Decided Upstream, Written Late.” arXiv 2608.08032, 2026.
33. E. V. Stein et al. “BabelSteering: Multilingual Safety Alignment via English Steering Vectors.” arXiv 2608.16577, 2026.
34. T. C. Addagada. “Structural Jailbreaks Generalize but Do Not Compound.” arXiv 2609.08373, 2026.
35. T. T. H. Nguyen et al. “Investigating the Influence of Prompt and Response Languages on LLM Content Generation.” arXiv 2608.26186, 2026.
36. J. Fiedler. “Bias and Uncertainty in LLM-as-a-Judge Estimation.” arXiv 2605.06939, 2026.
37. A. N. Angelopoulos et al. “Prediction-Powered Inference.” Science 382, 669–674, 2023.